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NCE Exam Practice Platform — Mauritius

2026

Information & Communication Technology

Grade 9 — National Certificate of Education (NCE)

PREDICTED PAPER 1 — Computer Systems, Binary/Hex & LAN Networks

Specimen Practice Paper — Not an Official MES Document

Duration: 1 hour 45 minutes | **Total Marks:** 100

Candidate Name: _____

Class / School: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name and class in the spaces provided above.
2. Answer **ALL** questions in all three sections.
3. Section A: Tick **ONE** box only for each multiple-choice question.
4. Section B: Write your answers in the spaces provided.
5. Section C: Use additional paper if the space provided is insufficient.
6. Calculators are **NOT** permitted.
7. Write clearly in black or blue ink; pencil may be used for diagrams only.

Section	Topic	Marks
A	Multiple Choice Questions	25
B	Short Structured Questions	40
C	Extended / Scenario Questions	35
TOTAL		100

SECTION A — Multiple Choice Questions [25 marks]

For each question, tick (✓) the box next to the **ONE** correct answer. Each question is worth **1 mark**.

1. Which component of the CPU is responsible for performing arithmetic and logical operations? 1
- ☐ A Control Unit (CU)
- ☐ B Arithmetic and Logic Unit (ALU)
- ☐ C Random Access Memory (RAM)
- ☐ D Register file
2. Which type of memory is *volatile* — it loses all data when the power is switched off? 1
- ☐ A ROM
- ☐ B RAM
- ☐ C Flash storage
- ☐ D DVD-ROM
3. The motherboard is best described as: 1
- ☐ A the main output device of a computer
- ☐ B a type of secondary storage
- ☐ C the main circuit board that connects all internal components
- ☐ D a device that processes graphics only
4. A CPU has a clock speed of 3.2 GHz. What does this mean? 1
- ☐ A It uses 3.2 gigabytes of RAM per second
- ☐ B It can execute 3.2 billion cycles per second
- ☐ C It transfers 3.2 gigabits of data per second
- ☐ D It runs at a temperature of 3.2 degrees
5. Which of the following is an example of *primary storage*? 1
- ☐ A External hard disk
- ☐ B USB flash drive
- ☐ C RAM
- ☐ D Blu-ray disc
6. Cache memory is placed close to the CPU in order to: 1

- ☐ A provide permanent storage for the operating system
- ☐ B store frequently accessed data for faster retrieval
- ☐ C replace RAM when it is full
- ☐ D cool the processor

7. A monitor and a speaker are both examples of:

1

- ☐ A input devices
- ☐ B storage devices
- ☐ C processing devices
- ☐ D output devices

8. Which of the following is a disadvantage of a Solid State Drive (SSD) compared to a Hard Disk Drive (HDD)?

1

- ☐ A SSDs are slower to access data
- ☐ B SSDs are generally more expensive per gigabyte
- ☐ C SSDs are larger and heavier
- ☐ D SSDs generate more heat

9. What is the decimal value of the binary number 1011_2 ?

1

- ☐ A 9
- ☐ B 10
- ☐ C 11
- ☐ D 12

10. Which of the following is the correct 8-bit binary representation of the decimal number 45?

1

- ☐ A 00101100
- ☐ B 00101101
- ☐ C 00110101
- ☐ D 00101111

11. How many bits are there in one byte?

1

- ☐ A 2
- ☐ B 4
- ☐ C 8
- ☐ D 16

12. The hexadecimal number $A6_{16}$ is equal to which decimal value?

1

- ☐ A 166

- ☐ B 160
- ☐ C 176
- ☐ D 156

13. The binary number $1111\ 1111_2$ expressed in hexadecimal is:

1

- ☐ A EE
- ☐ B FF
- ☐ C FE
- ☐ D EF

14. ASCII is a standard that:

1

- ☐ A compresses image files
- ☐ B assigns a numerical value to each character
- ☐ C encrypts network traffic
- ☐ D encodes video streams

15. What is the result of adding $0011\ 1100_2 + 0000\ 1010_2$?

1

- ☐ A 0100 0110
- ☐ B 0100 0100
- ☐ C 0011 1110
- ☐ D 0100 0111

16. One kilobyte (1 KB) is equal to:

1

- ☐ A 1000 bytes
- ☐ B 1024 bytes
- ☐ C 1024 bits
- ☐ D 1000 bits

17. LAN stands for:

1

- ☐ A Large Area Network
- ☐ B Linked Access Node
- ☐ C Local Area Network
- ☐ D Local Access Node

18. In a **bus topology**, all devices are connected to:

1

- ☐ A a central switch
- ☐ B each other in a loop
- ☐ C a single shared central cable (backbone)
- ☐ D every other device individually

19. What is the main function of a *router* in a network? 1
- ☐ A To amplify the network signal over long distances
 - ☐ B To direct data packets between different networks using IP addresses
 - ☐ C To convert digital signals into analogue for transmission
 - ☐ D To provide wireless access to a single device
20. HTTP stands for: 1
- ☐ A HyperText Transfer Protocol
 - ☐ B High Transfer Text Program
 - ☐ C HyperText Transfer Page
 - ☐ D High Text Transfer Protocol
21. In a star topology network, the device at the centre that all others connect to is typically a: 1
- ☐ A Repeater
 - ☐ B Modem
 - ☐ C Switch or Hub
 - ☐ D Bridge
22. What does IP stand for in the context of networking? 1
- ☐ A Internal Port
 - ☐ B Internet Protocol
 - ☐ C Integrated Program
 - ☐ D Interface Packet
23. A MAC address is: 1
- ☐ A a software-assigned network address that can be changed
 - ☐ B a unique hardware identifier assigned to a network interface card
 - ☐ C the IP address of a router
 - ☐ D a domain name used to locate a website
24. Wi-Fi transmits data using: 1
- ☐ A fibre optic cables
 - ☐ B electrical signals through copper wire
 - ☐ C wireless radio waves
 - ☐ D infrared light beams
25. Which network topology provides the highest fault tolerance because every device connects directly to every other device? 1

- ☐ A Bus
- ☐ B Star
- ☐ C Ring
- ☐ D Mesh

SECTION B — Short Structured Questions [40 marks]

Answer **ALL** questions. Write your answers in the spaces provided.

B1. CPU Components

5

- (a) State what *ALU* stands for.[1]

- (b) State the function of the *Control Unit (CU)* in the CPU.[1]

- (c) Explain what *clock speed* means and how a higher clock speed affects performance. [2]

- (d) State *one* difference between L1 cache and L2 cache.[1]

B2. Binary Conversion

5

- (a) Convert the binary number 10110101_2 to its decimal equivalent. Show your working. [2]

- (b) Convert the decimal number 179 to an 8-bit binary number. Show your working. [2]

- (c) Add the binary numbers 00110011_2 and 00101010_2 . Express your answer in binary. [1]

B3. Memory Types

5

- (a) State *two* differences between RAM and ROM.[2]

- (b) Give *one* example of data that is stored in ROM.[1]

- (c) Explain why a computer needs *both* RAM and secondary storage.[2]

B4. Storage Devices

5

- (a) Give *one* advantage of an SSD over an HDD.[1]

- (b) A school stores 500 student records, each 4 KB in size. Calculate the total storage required.
Give your answer in KB and in MB. [2]

- (c) Name *one* optical storage device.[1]

-
- (d) State *one* advantage of cloud storage compared to local secondary storage. [1]
-

B5. Network Topologies

5

- (a) In the box below, draw and label a *star topology* containing a central switch and *four* connected devices. [2]



- (b) State *one* advantage of a star topology over a bus topology. [1]
-

- (c) Name the topology in which every device connects directly to every other device. [1]
-

- (d) State *one* disadvantage of a ring topology. [1]
-

B6. Network Protocols

5

- (a) Explain the role of *TCP* (Transmission Control Protocol) in data communication. [2]
-
-
-

- (b) State the purpose of *DNS* (Domain Name System). [1]
-

(c) State the difference between *HTTP* and *HTTPS*.^[2]

B7. Hexadecimal

5

(a) Convert the hexadecimal number $3A_{16}$ to its decimal equivalent. Show your working. ^[2]

(b) Convert the decimal number 215 to hexadecimal. Show your working. ^[2]

(c) State *one* reason why programmers prefer to use hexadecimal rather than binary. ^[1]

B8. Input and Output Devices

5

(a) For each device below, write **INPUT** or **OUTPUT** in the table.^[2]

Device	Input or Output
Touchscreen	
Speakers	
Barcode reader	
Projector	

(b) Explain one difference between a 1080p monitor and a 4K monitor.^[2]

(c) Give *one* device that can function as **both** an input and an output device. ^[1]

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SECTION C — Extended / Scenario Questions [35 marks]

Answer **ALL** questions. Read each scenario carefully before answering.

C1. Scenario: School Computer Laboratory Network

12

Riverside College is equipping a new computer laboratory with 20 workstations. The ICT coordinator must design a network to connect all computers, share a printer, and provide internet access. She is choosing between a *bus topology* and a *star topology*. All computers will use Ethernet cables internally, and the internet connection will be provided by a fibre-optic line from the ISP.

- (a) In the space below, draw a labelled diagram of a **star topology** for the computer laboratory. Your diagram must show the central network device, at least four workstations, the printer, and the connection to a router. [3]



- (b) Explain **two reasons** why the ICT coordinator should choose a star topology rather than a bus topology for the laboratory. [4]

Reason 1:

Reason 2:

- (c) Name *one* piece of hardware needed to connect the laboratory network to the internet, and explain its function. [2]

- (d) The school assigns IP addresses in the range 192.168.1.1 to 192.168.1.30 to its devices. State what type of IP addressing this represents and give one advantage of this approach. [3]

C2. Binary and Hexadecimal — Colour Representation

10

Digital images use the *RGB colour model* where each pixel is defined by three values: Red, Green, and Blue. Each value is an integer from 0 to 255, stored as 8 bits. A pixel in a photograph has the RGB values: R = 200, G = 100, B = 50.

- (a) Convert **each** of the three RGB values to both 8-bit binary and hexadecimal. Complete the table below. [6]

Colour Channel	Decimal Value	8-bit Binary	Hexadecimal
Red (R)	200		
Green (G)	100		
Blue (B)	50		

- (b) Explain how a 24-bit colour system is able to represent more than 16 million different colours. [2]

- (c) A bitmap image measures 640×480 pixels and uses 24-bit colour depth. Calculate the uncompressed file size. Give your answer in **bytes** and in **megabytes** (to 2 decimal places). Show your working. [2]

C3. Scenario: Comparing Two Laptops

8

Two students are comparing their laptops:

Feature	Laptop A	Laptop B
Processor	Dual-core, 2.4 GHz	Quad-core, 3.6 GHz
RAM	4 GB	16 GB
Storage	5400 rpm HDD (500 GB)	SSD (512 GB)
Cache	2 MB L2 cache	8 MB L3 cache

- (a) State **three** reasons why Laptop B would be faster than Laptop A for general computing tasks. [3]

- (b) Explain what *quad-core* means and give *one* benefit for the user. [2]

- (c) A user runs a web browser, a word processor, and a video player simultaneously on Laptop A. Explain, with reference to RAM, why Laptop A may respond slowly. [3]

C4. Connecting to a Website

5

- (a) A student types **www.nouklass.com** into a web browser and presses Enter. Describe the sequence of events that occurs until the web page is displayed on screen. Your answer **must** include reference to: DNS, IP address, HTTP/HTTPS, and the server response. [5]

MARK SCHEME — Paper 1

This mark scheme is intended for practice purposes. Award marks for any equivalent correct answer unless a specific wording is required. For explanations, accept any response that demonstrates the correct understanding.

SECTION A — Multiple Choice Answers

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	C	4	B	5	C
6	B	7	D	8	B	9	C	10	B
11	C	12	A	13	B	14	B	15	A
16	B	17	C	18	C	19	B	20	A
21	C	22	B	23	B	24	C	25	D

SECTION B — Short Answers

Q	Expected Answer	Marks
B1(a)	Arithmetic and Logic Unit	1
B1(b)	Directs / coordinates the operation of all other parts of the CPU; fetches, decodes and executes instructions (any 1 correct point)	1
B1(c)	Clock speed = the number of cycles a CPU executes per second (measured in Hz/GHz) [1]; higher clock speed means more instructions per second / faster processing [1]	2
B1(d)	L1 is smaller but faster / built directly onto the processor core; L2 is larger but slightly slower / may be shared between cores (accept any correct difference)	1
B2(a)	$128+32+16+4+1 = 181$. [1 for correct method; 1 for correct answer: 181]	2
B2(b)	10110011 [1 for correct method; 1 for correct answer]	2
B2(c)	$00110011 + 00101010 = 01011101$ (93)	1
B3(a)	RAM is volatile / ROM is non-volatile [1]; RAM can be read and written / ROM is read-only (or written at manufacture) [1]	2
B3(b)	BIOS / firmware / boot instructions (any correct example)	1
B3(c)	RAM is needed for currently running programs (temporary) [1]; secondary storage is needed for permanent data when the computer is off [1]	2
B4(a)	Faster data access / no moving parts so more durable / silent / uses less power / smaller (any one)	1
B4(b)	$500 \times 4 = 2000$ KB [1]; $2000 \div 1024 \approx 1.95$ MB [1]	2
B4(c)	CD / DVD / Blu-ray (any one)	1

B4(d)	Accessible from any location with internet / automatic backup / easy to share files (any one)	1
B5(a)	Diagram showing central switch connected to 4 devices by individual lines, all labelled [1 for correct layout, 1 for labels]	2
B5(b)	If one cable fails only that device is affected (not the whole network) / easier to add/remove devices / easier to identify faults (any one)	1
B5(c)	Mesh topology	1
B5(d)	If one device or cable fails the entire ring is disrupted / slow / expensive to install (any one)	1
B6(a)	TCP breaks data into packets [1]; ensures all packets arrive and are reassembled in correct order, requests retransmission if lost [1]	2
B6(b)	Translates domain names (e.g. www.noukclass.com) into IP addresses	1
B6(c)	HTTP transmits data without encryption [1]; HTTPS encrypts data using SSL/TLS making it secure [1]	2
B7(a)	$3 \times 16 + 10 = 48 + 10 = 58$ [1 for method; 1 for correct answer: 58]	2
B7(b)	$215 \div 16 = 13$ remainder 7; so $D7_{16}$ [1 for method; 1 for D7]	2
B7(c)	Hex is more compact / easier to read than long binary strings; one hex digit represents exactly 4 bits (any correct reason)	1
B8(a)	Touchscreen: INPUT; Speakers: OUTPUT; Barcode reader: INPUT; Projector: OUTPUT [1 per 2 correct]	2
B8(b)	1080p has 1920×1080 pixels; 4K has 3840×2160 pixels [1]; 4K has approximately 4 times more pixels giving a sharper, higher resolution image [1]	2
B8(c)	Touchscreen / headset / disk drive / printer with scanner (any appropriate dual-function device)	1

SECTION C — Extended Answers

Q	Expected Answer / Mark Points	Marks
C1(a)	Diagram: central switch [1]; 4+ workstations connected with individual lines to switch [1]; printer and router also shown and labelled [1]	3
C1(b)	Per reason (max 2 marks each): fault tolerance — if one cable fails only one device is affected [1+1]; easier fault diagnosis — isolate which cable/device has a problem [1+1]; performance — dedicated connection to switch avoids collisions [1+1]; easy to add devices [1+1]	4
C1(c)	Router [1]; directs data packets between the school LAN and the internet using IP addresses [1]	2
C1(d)	Private / internal IP addressing (accept "static local IP") [1]; addresses do not conflict with the public internet [1]; cost-effective — no need to purchase public IP addresses for each device [1]	3

C2(a)	R=200: 11001000, C8 [2]; G=100: 01100100, 64 [2]; B=50: 00110010, 32 [2]	6
C2(b)	Each channel uses 8 bits, so $2^8=256$ values per channel [1]; three channels give $256 \times 256 \times 256 = 16,777,216$ combinations [1]	2
C2(c)	$640 \times 480 = 307,200$ pixels [1]; $\times 3$ bytes (24 bits) = 921,600 bytes; $\div (1024 \times 1024) = 0.88$ MB [1]	2
C3(a)	Any 3 from: higher clock speed (3.6 GHz vs 2.4 GHz); more cores (quad vs dual); more RAM (16 GB vs 4 GB); faster storage (SSD vs HDD) [1 each]	3
C3(b)	Quad-core means the CPU has four independent processing cores [1]; can run multiple tasks truly simultaneously / better multitasking performance [1]	2
C3(c)	Each running application needs to be held in RAM [1]; with only 4 GB, RAM becomes full when multiple apps run [1]; the OS must use the hard disk as virtual memory (page file/swap), which is much slower, causing lag [1]	3
C4(a)	Browser sends the domain name to a DNS server [1]; DNS resolves www.nouklass.com to an IP address and returns it [1]; browser sends an HTTP/HTTPS request to that IP address [1]; the web server processes the request and sends back the HTML/resources [1]; browser renders the page and displays it [1]	5